

Aditya Rao

Quantitative Developer (C++)

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Profile

Quantitative developer with 5 years building pricing and risk components in modern C++ at an options market maker on NSE. Own the Greeks pipeline end-to-end: model code, pybind11 API, regression suite, and the latency budget. Strong fixed-income and vol-surface mathematics. Notice period: 30 days.

Skills

C++: C++20, templates, cache-aware design, SIMD (AVX2), multithreading, perf/VTune profiling, CMake, Catch2

Python: pybind11 APIs, NumPy interop, pytest, tooling

Quant: G-Sec/OIS curve construction, vol surfaces (SVI), Greeks via AAD (co-developed in-house tape), Monte Carlo and PDE pricers, QuantLib (contributor: 2 merged PRs)

Platform: Linux, Git, Jenkins, Grafana

Experience

Quantitative Developer — Photon Trading Systems, Bengaluru

Jun 2021 – present

- Own the intraday Greeks engine (C++20) for the NSE index-options desk: 1.4M instrument revals/second on a fixed 32-core budget.
- Co-developed the firm's AAD tape, replacing bump-and-revalue for vega/vanna; nightly risk run cut from 3.2h to 41min.
- Maintain the pybind11 research API used daily by 12 quants; zero breaking releases in 3 years (semantic-versioned, golden-number gated).
- Reduced pricing-library build time 55% (modular headers, precompiled modules), directly speeding desk hotfix turnaround.

Software Engineer (C++) — Samsung R&D Institute, Bengaluru

Jul 2019 – May 2021

- On-device multimedia pipeline performance work (C++17): 30% latency reduction on the camera hot path.
- Introduced sanitizer-clean CI (ASan/UBSan) for the module.

Education

B.Tech Computer Science & Engineering — NITK Surathkal, 2015 – 2019

Open source

QuantLib: day-count convention fix + SVI calibration test coverage (merged 2024).

Languages known

Kannada (native), English (fluent), Hindi (working), Konkani (working)